3.8 Polynomial Rings in One Variable

Definition 3.8.1 (Polynomial Ring)
Let A be a commutative ring. Define the polynomial ring $A[X]$ to be the monoid ring $A[\mathbb{N}]$ consisting of formal

$$f(X) = \sum_{i=0}^{\infty} a_i X^i$$

such that only finitely many a_i are non-zero. Define degree in the obvious way

$$\deg(f) = \inf\{n \mid m > n \implies a_m = 0\} < \infty$$

and the leading coefficient to be $\ell(f) := a_{\deg(f)}$. By convention $\deg(0) = -\infty$.

Definition 3.8.2 (Monic polynomial)
*Let $f \in A[X]$. We say f is **monic** if the leading coefficient, $\ell(f)$, is 1.*

Lemma 3.8.3
If A is an integral domain then for elements $f, g \in A[X]$

$$\deg(fg) = \deg(f) + \deg(g)$$

$$\ell(fg) = \ell(f)\ell(g)$$

Further $A[X]$ is an integral domain.

Proposition 3.8.4 (Nilpotent and Invertible Polynomials)
Let A be a ring then

a) $N(A[X]) = N(A)[X] \subset A[X]$
b) $A[X]^\star = A^\star + XN(A)[X]$

Proof. Suppose $a \in N(A)$ then clearly aX^i is nilpotent. Therefore $N(A)[X] \subseteq N(A[X])$ since the nilradical is an ideal. Conversely suppose $f \in A[X]$ is nilpotent, i.e. $f^n = 0$. For any prime ideal $\mathfrak{p} \lhd A$ we find that $\overline{f}^n = 0$ as an element of $(A/\mathfrak{p})[X]$. As $A/\mathfrak{p}$ is an integral domain we have by the previous Lemma $\overline{f} = 0$. As $\mathfrak{p}$ was arbitrary and $N(A) = \bigcap \mathfrak{p}$ we see that $f \in N(A)[X]$ as required.

Suppose $f \in A[X]^\star$ and $fg = 1$, then clearly the constant term of f must be invertible. Reduce modulo $\mathfrak{p}$ to find $\deg\left(\overline{f}\right) + \deg(\overline{g}) = 0 \implies \deg\left(\overline{f}\right) = \deg(\overline{g}) = 0$, which means $\overline{f}$ is a constant polynomial. As $\mathfrak{p}$ was arbitrary we see again that the other coefficients of f must be nilpotent as required. Therefore $A[X]^\star \subseteq A^\star + XN(A)[X]$. Conversely by (3.3.67) and a) we have $A^\star + XN(A)[X] \subseteq A[X]^\star + N(A[X]) \subseteq A[X]^\star$ so the result follows. ∎

It satisfies the following universal property

Proposition 3.8.5 (Evaluation at a point)
Consider an A-algebra B and $b \in B$. Then there exists a unique A-algebra homomorphism

$$\mathrm{ev}_b : A[X] \to B$$

such that $\mathrm{ev}_b(X) = b$. We write $p(b) = \mathrm{ev}_b(p)$. It is given by

$$p(b) = \sum_{k=0}^{\deg(p)} i_B(a_k) b^k$$

The image of ev_p is equal to $A[b]$ the smallest sub-A-algebra generated by b. For any morphism $\phi : B \to C$ such that $\phi(b) = c$ we have

$$\phi \circ \mathrm{ev}_b = \mathrm{ev}_c$$

Remark 3.8.6
In categorical jargon $A[X]$ is an initial object in the category of pointed A-algebras.

Proposition 3.8.7 (Base Change of the Polynomial Ring)
Let $\phi : A \to B$ be a homomorphism of commutative rings. Then there is a natural isomorphism of B-algebras

$$A[X]_{(B)} \xrightarrow{\sim} B[X]$$
$$b \otimes \sum_i a_i X^i \to \sum_i b\phi(a_i)X^i$$

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

For any B-algebra Z (in particular an A-algebra via ϕ), combining (3.4.38) with the universal property of $A[X]$ (3.8.5) (applied to the underlying A-algebra of Z) gives a bijection natural in Z

$$\mathrm{AlgHom}_B\big(A[X]_{(B)}, Z\big) \cong \mathrm{AlgHom}_A(A[X], Z) \cong Z, \quad \psi \to \psi(1 \otimes X)$$

which is exactly the universal property of $B[X]$ (3.8.5) (applied with base ring B); by the Yoneda Lemma (2.6.58), this yields the required isomorphism (the image of $1_{A[X]_{(B)}}$ under it, sending $X \to 1 \otimes X$). ∎

Proposition 3.8.8 (Evaluation commutes with algebra homomorphism)
Let $\phi : B \to C$ be a homomorphism of A-algebras and $p \in A[X]$ then

$$\phi(p(b)) = p(\phi(b))$$

Proposition 3.8.9 (Formal Derivative)
Let A be a ring. Then there exists a unique A-module homomorphism

$$(-)' : A[X] \to A[X]$$

such that

$$(X^r)' = \begin{cases} 0 & r = 0 \\ rX^{r-1} & r > 0 \end{cases}$$

It satisfies the product rule

$$(fg)' = f'g + fg'$$

for all $f, g \in A[X]$. Define recursively $f^{(0)} = f$ and $f^{(n)} := f^{(n-1)'}$.

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

By (3.8.1), $A[X]$ is the free A-module on $\{1, X, X^2, ...\}$, so $(-)'$ exists and is unique.

Suppose $f(X) = \sum_{i=0}^n a_i X^i$ and $g(X) = \sum_{i=0}^m b_i X^i$. Then

$$(fg)(X) = \sum_{i=0}^{n+m} \left(\sum_{k+l=i} a_k b_l \right) X^i$$

and

$$\begin{aligned}
f'(X) &= \sum_{i=0}^{n-1} (i+1) a_{i+1} X^i \\
g'(X) &= \sum_{i=0}^{m-1} (i+1) b_{i+1} X^i \\
(fg)'(X) &= \sum_{i=0}^{n+m-1} (i+1) \left(\sum_{k+l=i+1} a_k b_l \right) X^i \\
&= \sum_{i=0}^{n+m-1} \left(\sum_{k+l=i+1} (k+l) a_k b_l \right) X^i \\
&+ \sum_{i=0}^{n+m-1} \left(\sum_{k+l=i} (k+1) a_{k+1} b_l \right) X^i + \sum_{i=0}^{n+m-1} \left(\sum_{k+l=i} (l+1) a_k b_{l+1} \right) X^i \\
&= f'(X)g(X) + f(X)g'(X)
\end{aligned}$$

∎

Corollary 3.8.10 (Taylor Expansion mod h^2)
Let A be a ring, $f \in A[X]$, and $a, h \in A$. Then

$$f(a+h) - f(a) - f'(a)h \in (h^2) \triangleleft A.$$

Proof. Drafted by Claude Sonnet 5, reviewed by David Rufino.

Write $f(X) = \sum_{i=0}^{d} c_i X^i$. We first show, by induction on $i \geq 0$, that $(a+h)^i - a^i - ia^{i-1}h \in (h^2)$ (reading $a^{-1}h := 0$ for $i = 0$). The cases $i = 0, 1$ are immediate equalities ($1 - 1 - 0 = 0$, $(a+h) - a - h = 0$). Given $(a+h)^i = a^i + ia^{i-1}h + h^2 z$ for some $z \in A$:

$$(a+h)^{i+1} = (a+h)(a^i + ia^{i-1}h + h^2 z) = a^{i+1} + (i+1)a^i h + h^2(az + ia^{i-1} + hz),$$

so $(a+h)^{i+1} - a^{i+1} - (i+1)a^i h \in (h^2)$, completing the induction. Multiplying the case-i congruence by c_i and summing over $i = 0, ..., d$ (as (h^2) is an ideal) gives $f(a+h) - f(a) - f'(a)h \in (h^2)$, noting $f'(a) = \sum_i c_i \cdot ia^{i-1}$ by (3.8.9) evaluated at a (3.8.5). ∎

Lemma 3.8.11 (Shifted Powers Form a Basis)
Let k be a field, $\theta \in k$, and $n \geq 1$ an integer. Then $1, (X-\theta), ..., (X-\theta)^{n-1}$ is a k-basis for the space of polynomials in $k[X]$ of degree $< n$.

Proof. Using (3.8.5) we may apply an isomorphism $X \to X - \theta$, from which the result follows immediately. ∎

Definition 3.8.12 (Conjugate polynomial)
Let $\phi : A \to B$ be a homomorphism and $f \in A[X]$, then define

$$f^{\phi}(X) := \sum_{i=0}^{n} \phi(a_i) X^i$$

It induces a ring homomorphism

$$A[X] \to B[X]$$

and has the property that

$$f^{\phi}(\phi(a)) = \phi(f(a))$$

Proposition 3.8.13 (Division Algorithm I)
Let A be an integral domain and $f(X) \in A[X]$ a polynomial and $g(X) \in A[X]$ a non-zero monic polynomial. Then there exists unique polynomials $q(X)$ and $r(X)$ such that

- $f(X) = q(X)g(X) + r(X)$
- $\deg(r) < \deg(g)$

In particular when $\deg(g) = 1$ then $r \in A$.

Proof. If $\deg(f) < \deg(g)$ then $q = 0$ and $r = f$. Otherwise assume $n = \deg(f) \geq \deg(g) = m$ and proceed by induction on n. Note that since g is monic then we have $f - \ell(f)gX^{n-m}$ has degree $n-1$, so by induction

$$f - \ell(f)gX^{n-m} = q'g + r$$

with $\deg(r) < \deg(g)$. Therefore

$$f = (q' + \ell(f)X^{n-m})g + r$$

as required.

Suppose $qg + r = q' + r'$ then $(q - q')g = (r' - r)$. This implies that $\deg((q-q')g) < \deg(g)$ which implies $(q-q')g = 0 \implies q = q'$ and $r = r'$. This demonstrates uniqueness. ∎

Proposition 3.8.14
Let B be an A-algebra and $f \in B$ not nilpotent. Then there is a canonical isomorphism

$$B[X]/(1 - Xf) \to B_f$$
$$\overline{P(X)} \to P(1/f)$$

In particular if B is a finitely-generated A-algebra, then so is B_f.

Proof. By (3.8.5) there is a A-algebra homomorphism $\phi : B[X] \to B_f$ such that $\phi(X) = \frac{1}{f}$. Evidently $(1 - Xf) \subset \ker(\phi)$ so by (...) this induces an A-algebra homomorphism $B'_f := B[X]/(1-Xf) \to B_f$. Evidently $\overline{X} = \overline{f}^{-1}$ in B'_f, so

by (3.6.5) there is a map $\psi : B_f \to B[X]/(1 - Xf)$ given by $\frac{b}{f^r} \to \overline{bX^r}$. One may verify that ψ is a two-sided inverse for ϕ, and so is an isormophism.

The final statement follows from (3.3.75). ■

Corollary 3.8.15
For k a field, the units of $k[X]$ are precisely the non-zero polynomials of degree 0.

Proof. As k is a field, $N(k) = 0$, so by (3.8.4) $k[X]^\star = k^\star + X \cdot 0 = k^\star$, that is to say the non-zero constants. ■

Proposition 3.8.16 (Division Algorithm over a field)
For k a field consider the polynomial ring $k[X]$. For every pair of polynomials $f(X)$, $g(X)$ there exists unique polynomials $q(X)$ and $r(X)$ such that

$$f(X) = q(X)g(X) + r(X)$$

and $\deg(r) < \deg(g)$.

Proof. Apply (3.8.13) to $g/\ell(g)$, and multiply by $\ell(g)$ again. ■